

CHARLESTON INTL, SC, USA

WMO: 722080

Lat: 32.8950N

Lon: 80.0275W

Elev: 14

StdP: 101.15

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13880

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.7	-0.8	-12.9	1.2	2.3	-10.1	1.6	3.8	11.4	16.6	10.2	14.7	2.6	30	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	34.7	25.4	33.5	25.2	32.4	24.9	26.9	31.5	26.5	31.0	26.0	30.2	4.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.2	29.0	25.3	20.4	28.5	24.8	19.9	28.1	84.8	31.7	82.3	30.8	80.2	30.3	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.2	7.4	DB	-6.3	36.8	1.8	1.1	-7.6	37.6	-8.6	38.3	-9.6	38.9	-10.9	39.7	(4)
(5)			WB	-7.7	28.2	1.7	0.6	-8.9	28.6	-9.9	29.0	-10.9	29.4	-12.1	29.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	19.3	9.7	11.5	14.9	18.9	23.0	26.5	27.9	27.4	24.9	19.9	14.5	11.5	(6)
(7)		DBStd	7.48	5.44	4.87	4.65	3.80	3.16	2.34	1.67	1.83	2.47	3.92	4.55	5.15	(7)
(8)		HDD10.0	179	74	36	13	1	0	0	0	0	0	0	11	44	(8)
(9)		HDD18.3	1012	270	194	124	38	4	0	0	1	30	130	219		(9)
(10)		CDD10.0	3558	64	79	164	269	403	494	554	539	447	308	147	89	(10)
(11)		CDD18.3	1349	2	4	17	57	149	244	296	281	197	79	16	6	(11)
(12)		CDH23.3	11973	10	27	122	402	1186	2282	3072	2749	1518	505	77	22	(12)
(13)	CDH26.7	4362	0	1	16	84	384	916	1289	1080	488	98	5	0	(13)	
(14)	Wind	WSAvg	3.5	3.6	3.7	4.0	4.0	3.7	3.4	3.2	3.0	3.3	3.1	3.2	3.3	(14)
(15)	Precipitation	PrecAvg	1251	86	82	91	75	73	141	152	167	134	106	67	77	(15)
(16)		PrecMax	1664	203	264	157	159	162	301	251	272	293	287	201	223	(16)
(17)		PrecMin	853	10	32	11	25	9	70	58	89	10	0	1	16	(17)
(18)		PrecStd	198	52	53	40	39	44	64	51	60	68	78	49	42	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.8	26.2	28.7	31.0	34.2	35.8	36.1	35.8	33.4	30.7	27.4	25.7	(19)
(20)			MCWB	18.7	18.7	18.8	20.8	23.1	25.3	25.8	26.3	24.4	23.6	21.3	20.1	(20)
(21)		2%	DB	22.4	23.9	26.3	28.9	32.1	34.2	34.6	33.9	31.8	29.0	25.5	23.6	(21)
(22)			MCWB	17.5	17.3	18.1	20.0	22.5	24.8	25.8	24.2	22.5	19.7	18.9	(22)	
(23)		5%	DB	20.3	21.7	24.5	27.4	30.3	32.7	33.4	32.7	30.7	27.7	23.9	21.6	(23)
(24)			MCWB	16.4	15.8	17.1	19.5	22.2	24.6	25.7	25.6	23.9	21.7	18.9	18.5	(24)
(25)		10%	DB	18.3	19.6	22.5	25.8	28.8	31.3	32.2	31.6	29.6	26.3	22.3	19.7	(25)
(26)			MCWB	15.0	15.2	16.2	18.7	21.6	24.2	25.5	25.2	23.5	20.9	18.3	16.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.4	20.5	20.9	23.2	25.1	26.9	27.9	27.9	26.3	25.4	22.8	21.7	(27)
(28)			MCD B	22.7	24.1	25.1	27.5	30.5	32.7	32.8	32.6	29.8	27.7	24.8	23.9	(28)
(29)		2%	WB	18.8	19.0	19.9	21.9	24.1	26.2	27.1	27.0	25.6	24.4	21.5	20.2	(29)
(30)			MCD B	21.2	22.1	23.9	26.1	28.7	31.5	31.8	31.2	29.2	27.1	23.7	22.2	(30)
(31)		5%	WB	17.4	17.7	18.8	21.0	23.5	25.6	26.6	26.5	25.1	23.5	20.4	18.9	(31)
(32)			MCD B	19.7	20.4	22.3	24.9	27.8	30.4	31.2	30.7	28.6	26.2	22.8	21.2	(32)
(33)		10%	WB	15.7	16.1	17.8	20.2	22.8	25.1	26.0	25.9	24.6	22.5	19.0	17.4	(33)
(34)			MCD B	17.8	18.7	21.1	23.9	26.9	29.5	30.4	29.9	27.8	24.9	21.5	19.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.2	11.5	11.7	11.6	10.5	9.3	8.9	8.5	8.7	10.4	11.6	10.9	(35)
(36)			MCDBR	12.4	13.1	13.2	12.3	11.6	10.7	10.0	9.6	9.6	10.6	12.0	11.4	(36)
(37)			MCWBR	8.3	7.7	6.4	5.3	4.3	4.0	3.8	3.6	3.6	5.0	6.9	7.5	(37)
(38)		5% WB	MCD BR	10.8	11.3	11.1	10.5	9.9	9.6	9.2	8.9	8.1	8.3	9.7	10.4	(38)
(39)			MCWBR	8.4	7.9	6.5	5.4	4.2	4.0	3.9	3.7	3.6	4.7	6.6	7.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.326	0.334	0.365	0.408	0.468	0.499	0.535	0.515	0.436	0.390	0.348	0.331	(40)	
(41)		taud	2.535	2.509	2.441	2.303	2.184	2.146	2.058	2.112	2.329	2.434	2.508	2.549	(41)	
(42)		Ebn at Noon	894	921	913	882	829	800	768	778	831	847	859	867	(42)	
(43)		Edh at Noon	84	94	109	130	148	154	167	156	121	101	85	79	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.85	3.51	4.64	5.93	6.40	6.50	6.24	5.63	4.79	4.12	3.17	2.53	(44)	
(45)		RadStd	0.16	0.36	0.42	0.40	0.43	0.45	0.38	0.35	0.40	0.45	0.23	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+118	(46)	
(47)	Regional (43 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+206	+144	(47)	

Nomenclature: See separate page